MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC TEST – 01 (POLYNOMIALS)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM:** 30

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 11 and Question 12(iii).
2. **Section A** comprises 5 questions (4 MCQs and 1 Assertion-Reason) of 1 mark each.
3. **Section B** comprises 2 VSAQ questions of 2 marks each.
4. **Section C** comprises 2 SA questions of 3 marks each.
5. **Section D** comprises 2 LA / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case Study with 3 sub-parts of 5 marks total.

SECTION A: Objective & MCQs (5 × 1 = 5 Marks)

Q1. If α, β are zeroes of $f(x) = ax^2 + bx + c$, then $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ is equal to: [1]

(A) $\frac{b^2 - 2ac}{c^2}$ (B) $\frac{b^2 - 2ac}{a^2}$ (C) $\frac{b^2 + 2ac}{c^2}$ (D) $\frac{b^2 - 2ac}{b^2}$

Q2. What should be added to the polynomial $x^2 - 5x + 4$ so that 3 is a zero of the resulting polynomial? [1]

(A) 1 (B) 2 (C) 4 (D) 5

Q3. If 2 and $-\frac{1}{2}$ are zeroes of the polynomial $px^2 + 5x + r$, then: [1]

(A) $p = r = 2$ (B) $p = r = -2$ (C) $p = 2, r = -2$ (D) $p = -2, r = 2$

Q4. If α, β are the zeroes of the polynomial $x^2 - 1$, then the value of $(\alpha + \beta)$ is: [1]

(A) 2 (B) -2 (C) 0 (D) 1

Q5. **Assertion (A):** The polynomial $p(x) = x^2 + 4x + 5$ has no real zeroes.
Reason (R): A quadratic polynomial $ax^2 + bx + c$ has no real zeroes if $b^2 - 4ac < 0$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. If the graph of a quadratic polynomial $ax^2 + bx + c$ cuts the positive direction of the y-axis, what is the sign of c? Justify your answer. [2]

Q7. Show that the quadratic polynomial $f(x) = x^2 + 2x + 5$ has no real zeroes. [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. If α and β are the zeroes of the polynomial $f(x) = x^2 + x - 2$, find the value of $\frac{1}{\alpha} - \frac{1}{\beta}$. [3]

Q9. If α, β are the zeroes of $f(x) = x^2 - 2x + 3$, find a quadratic polynomial whose zeroes are $\alpha + 2$ and $\beta + 2$. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. If α and β are the zeroes of the quadratic polynomial $f(x) = ax^2 + bx + c$, then evaluate: [5]

(i) $\alpha^4 + \beta^4$ (ii) $\frac{1}{a\alpha + b} + \frac{1}{a\beta + b}$

Q11. Find the zeroes of the quadratic polynomial $g(x) = a(x^2 + 1) - x(a^2 + 1)$, where $a \neq 0$. Verify the relationship between the zeroes and the coefficients. [5]

SECTION E: Applied Case Study Analysis (1 × 5 = 5 Marks)

Question 12: Case Study Analysis — Roller Coaster Track Geometry

An amusement park engineer designs the vertical drop section of a junior roller coaster. The drop and climb follow a parabolic curve modelled by the polynomial:

$$p(x) = \frac{1}{4}(x^2 - 6x + 5)$$

where x is horizontal distance (in tens of metres) and $p(x)$ is elevation relative to platform level (in tens of metres).

(i) Find the zeroes of the polynomial $p(x)$ and state their physical significance. [1]

(ii) Find the coordinates of the lowest dip point of the track and state the maximum drop depth. [2]

(iii) **(a)** If a safety sensor is fixed at $x = 3$, determine the elevation of the track at the sensor. [2]

— OR —

(b) If a modified track profile has zeroes at $x = 2$ and $x = 8$, write its quadratic polynomial equation. [2]

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